Description of the Container Closure System

Tenecteplase	**Global ID_Version** **206893_2808118_3.0**
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Tenecteplase drug substance is stored in 120 L mobile stainless steel cryo vessels of the steel quality 1.4435 / 316 L. This container closure system protects tenecteplase from microbial ingress as well as from degradation by light.

Each vessel is equipped with a 0.2 µm vent filter and a 0.2 µm liquid filter membrane. The outlet and inlet ports can be closed with a membrane valve.

The stainless-steel cryo vessels are cleaned-in-place (CIP), steamed-in-place (SIP) and integrity tested via a pressure hold test prior to the filtration of the drug substance into the vessel. For tank cleaning, CIP sprayballs are connected to the CIP inlet and cleaning solutions are directed into the tank through the CIP inlet and CIP sprayballs.

The mobile stainless steel cryo vessels, designed for freezing, thawing, and frozen storage of drug substance, have a heat transfer jacket and a coil assembly for the circulation of the heat transfer fluid (HTF) during the freezing or thawing operation (see Figure 1). The drug substance temperature during freezing or thawing is measured with an inserted resistance temperature detector (RTD).

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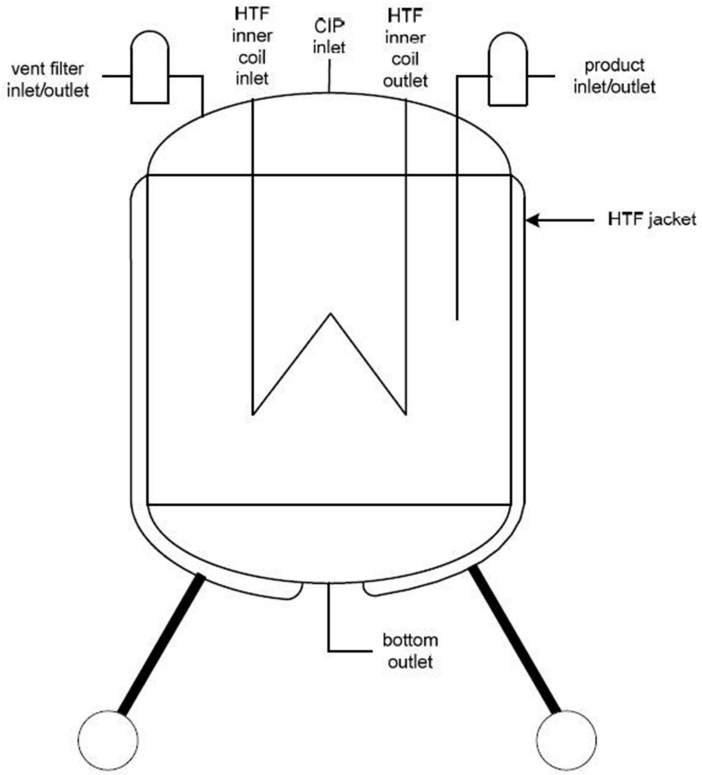


Figure 1 Schematic drawing of a stainless steel cryovessel

The stainless steel cryo vessel is the primary packaging component, since it is in direct contact with the drug substance. The vent filter membrane, inlet and outlet seals, and diaphragm are functional secondary packaging components, since they provide protection from microbial ingress.

Each container is individually labelled with compound name, batch number, date of filling, and filling volume.

The container closure components are described in Table 1.

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Table 1 Description of container closure system

Component	Material of construction
Stainless steel cryo vessel (primary)	1.4435 / 316 L Stainless steel
Vent filter membrane (secondary)	PVDF (Polyvinylidene Fluoride)
Inlet and outlet seals and diaphragm (secondary)	EPDM (Ethylene Propylene Diene Monomer)